Event Logging and Monitoring

Topic: Event Logging and Monitoring Lab
Task: Generate Windows Event Log and Monitor them

By **Kunal Jain**

1. Objective

The objective of this lab is to generate, locate, and analyze important Windows Security and System events on a Windows Virtual Machine (VM) using Event Viewer. The lab demonstrates authentication failures, successful logons, user account creation, administrator group membership changes, service creation and deletion, and security log clearing.

2. Lab Environment

- Operating System: Windows 10/11 or Windows Server VM
- Tool: Windows Event Viewer
- Log Sources: Windows Logs → Security and Windows Logs → System
- Administrative Privileges: Required for several activities
- Event Filtering: Filter Current Log using the relevant Event IDs

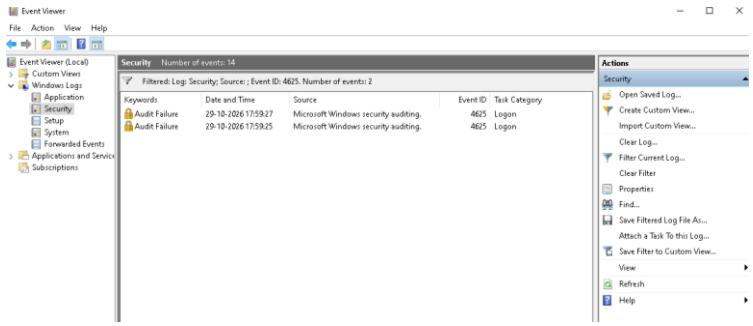
3. Procedure

Event 1: Failed Logon - Event ID 4625

Objective: Generate and identify an unsuccessful Windows logon attempt.

Procedure:

1. Lock the VM using Win + L, or sign out and return to the sign-in screen.
2. Attempt to sign in using an intentionally incorrect password for an existing test account.
3. Sign in with the correct credentials afterward, if required.
4. Open Event Viewer by running eventvwr.msc.
5. Navigate to Windows Logs → Security.
6. Select Filter Current Log from the Actions pane.
7. Enter 4625 in the Event IDs field and apply the filter.
8. Open the relevant event and inspect its details.



Event 2: Successful Logon - Event ID 4624

Objective: Identify a successful Windows logon and determine its logon type.

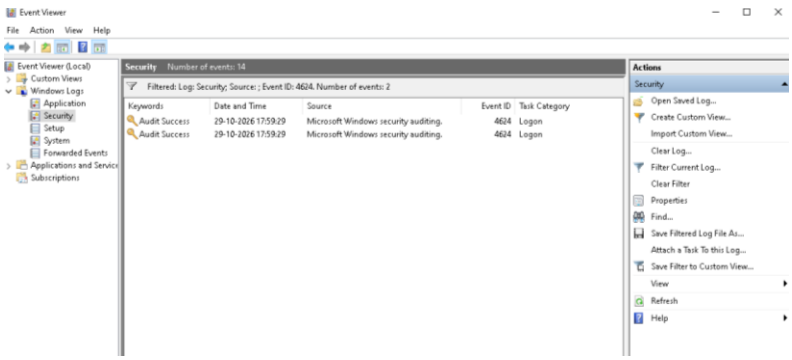
Procedure:

1. Sign in to the Windows VM using valid credentials.
2. Open Event Viewer.
3. Navigate to Windows Logs → Security.
4. Select Filter Current Log.

5. Enter 4624 in the Event IDs field and apply the filter.
6. Open a recent event corresponding to your test account.
7. In the General tab, locate Logon Type.

Common Logon Types:

- Type 2 - Interactive logon at the local computer
- Type 3 - Network logon
- Type 4 - Batch logon
- Type 5 - Service logon
- Type 7 - Workstation unlock
- Type 10 - Remote Interactive logon, typically Remote Desktop



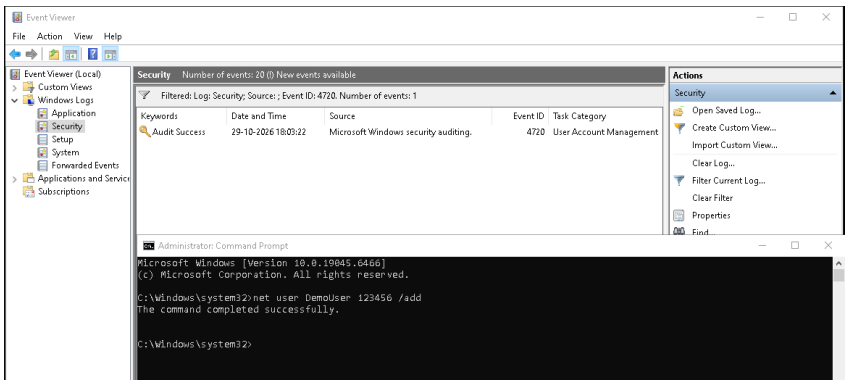
Event 3: New User Account - Event ID 4720

Objective: Create a test user and verify that Windows records the account creation.

Procedure:

1. Open Command Prompt or Windows Terminal as Administrator.
2. Create a lab-only local account:

   net user DemoUser 123456 /add
3. Enter a suitable password when prompted, following the VM's password policy.
4. Open Event Viewer and navigate to Windows Logs → Security.
5. Select Filter Current Log and enter 4720.
6. Open the event associated with the newly created account.

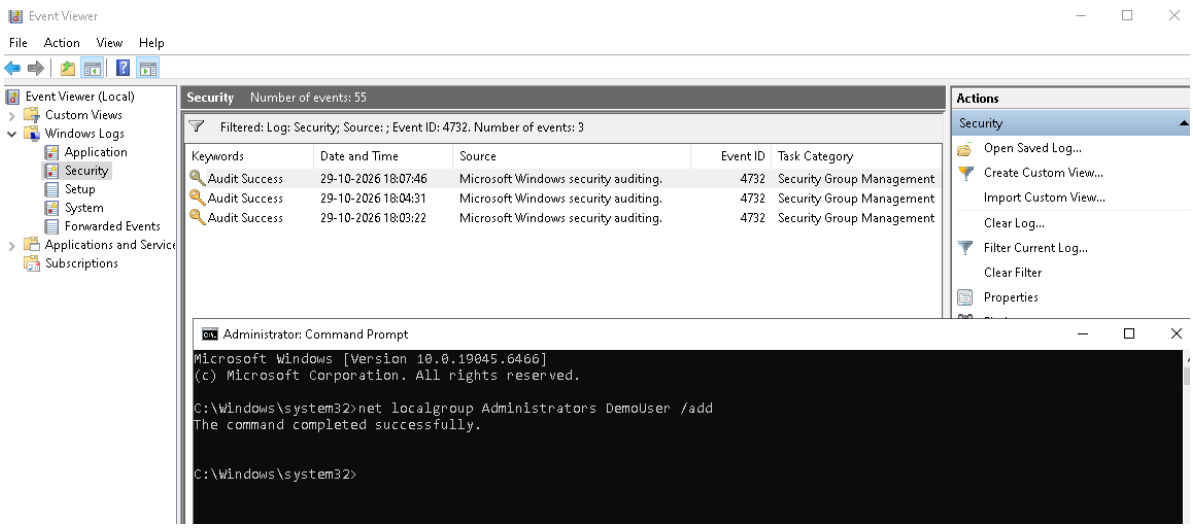


Event 4: User Added to Administrators - Event ID 4732

Objective: Verify that Windows records a user being added to the local Administrators group.

Procedure:

1. Ensure that DemoUser exists.
2. From an elevated Command Prompt, run:
   net localgroup Administrators DemoUser /add
3. Open Event Viewer.
4. Navigate to Windows Logs → Security.
5. Select Filter Current Log and enter 4732.
6. Open the event associated with the group membership change.



Note: Perform this action only in an authorized lab VM. The account will have elevated privileges until removed.

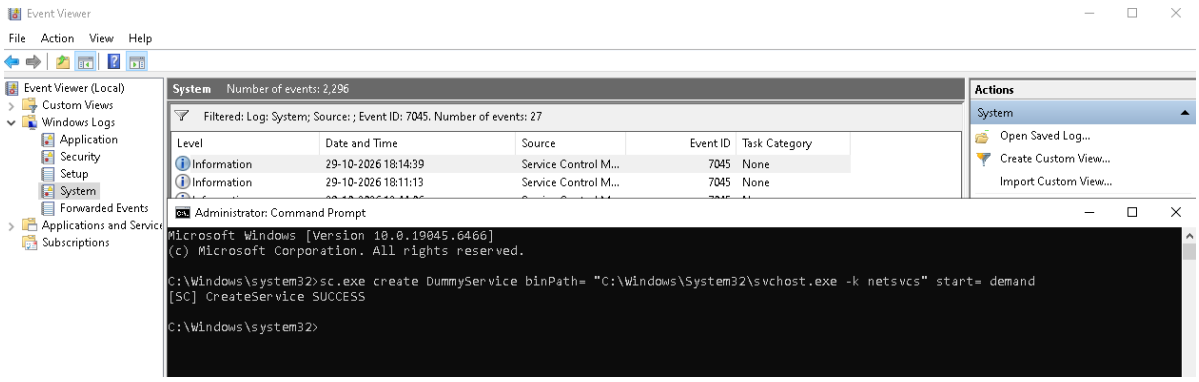
Event 5: Dummy Service Creation and Deletion - Event ID 7045

Objective: Create and delete a harmless test service and inspect the corresponding System log records.

Procedure:

1. Open Command Prompt as Administrator.
2. Create a dummy service that runs the built-in Windows command interpreter:
   sc.exe create DummyService binPath= "C:\Windows\System32\cmd.exe" start= demand
3. Verify the service creation response.
4. Open Event Viewer and navigate to Windows Logs → System.
5. Select Filter Current Log and enter 7045.
6. Locate the service installation event and inspect its details.
7. Return to the elevated Command Prompt and delete the test service:
   sc.exe delete DummyService

8. Refresh the System log and check for a corresponding service deletion event, if one is recorded.



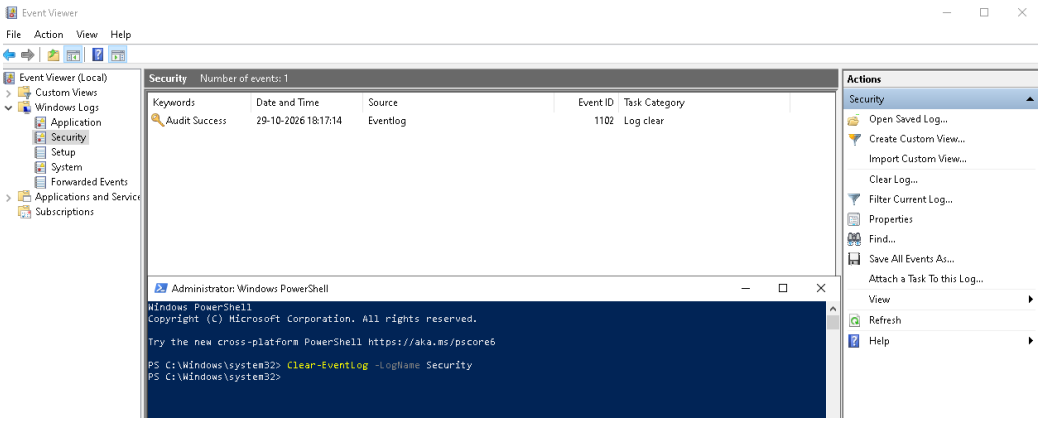
Important Clarification: Event ID 7045 records a service installation. Service deletion is not reliably recorded as Event ID 7045.

Event 6: Security Log Cleared - Event ID 1102

Objective: Understand the audit event generated when the Windows Security audit log is cleared.

Procedure:

1. Warning: Clearing the Security log removes existing audit records. Perform this only on a disposable, authorized lab VM after capturing all required earlier evidence.
2. Open Event Viewer as Administrator.
3. Navigate to Windows Logs → Security.
4. Choose Clear Log from the Actions pane.
5. If you need to preserve the existing records, select Save and Clear and store the exported log securely before proceeding. Otherwise, use the appropriate clear option for your lab.
6. Confirm the operation.
7. Return to the Security log and refresh the view.
8. Select Filter Current Log and enter 1102.
9. Inspect the event if it is present in the newly generated log.



Important Clarification: Event ID 1102 is generated in the Security log when the audit log is cleared. Because clearing the log removes previous records, collect screenshots of all earlier events first.

4. Results and Analysis

The lab demonstrates how Windows records important security and system activities through event logging.

Activity	Event ID	Log
Failed logon	4625	Security
Successful logon	4624	Security
User account created	4720	Security
Member added to local security group	4732	Security
Service installed	7045	System
Security audit log cleared	1102	Security

These events are useful for security monitoring, auditing, incident investigation, account management, and identifying potentially suspicious administrative activity.

5. Conclusion

This practical exercise demonstrates how to generate and investigate Windows authentication, account management, group membership, service installation, and audit log clearing events. Using Event Viewer and the Filter Current Log feature allows administrators and security analysts to locate specific events efficiently. The collected screenshots provide evidence of each activity and support further analysis of Windows security auditing.

Submission Note: Replace or supplement the expected results with the actual event details and screenshots collected from your own VM. Event availability depends on the Windows version, audit policy, permissions, and system configuration.